

Problem-Solution Fit

Date	06 July 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience.

1. CUSTOMER SEGMENT(S) [CS]	6. CUSTOMER LIMITATIONS [CL]	5. AVAILABLE SOLUTIONS PROS & CONS [AS]
- Bank Credit Officers: Frontline staff evaluating and approving individual loan applications. - Financial Analysts: Professionals conducting batch risk assessments during high-volume periods. - Banking / NBFC Risk Teams: Teams responsible for minimizing non-performing assets (NPAs).	- Technical Skills: Credit officers often lack programming or data-science skills to build risk models themselves. - Budget: Limited funds for expensive, enterprise-grade credit scoring or BI software. - Devices/Connectivity: Reliance on standard branch desktops with intermittent connectivity in some regions.	Current Solutions: Manual document review, credit bureau scores, static rule-based checklists. Pros: Regulatory acceptance, incorporates human judgement, well understood by staff. Cons: Slow, inconsistent across officers, doesn't scale during high application volume, prone to subjective bias.
2. PROBLEMS / PAINS + FREQUENCY [PR]	9. PROBLEM ROOT / CAUSE [RC]	7. BEHAVIOR + INTENSITY [BE]

• Pain: Inability to quickly and consistently assess repayment risk for applicants with irregular self-employment income or no credit history. • Pain: Manual review of every application is tedious and prone to human error. • Frequency: High – occurs with every loan application, and intensifies during high-volume periods.	The core issue is the lack of an accessible, real-time predictive tool that translates structured applicant data into an actionable approval/risk decision, without requiring advanced data-science expertise from credit staff.	• Behavior: Officers manually collect applicant documents, cross-check credit history, and apply informal rules of thumb to decide approval. • Intensity: High effort, high time investment, low agility – officers are stuck in a repetitive, manual evaluation loop.
--	---	---

3. TRIGGERS TO ACT [TR]	10. YOUR SOLUTION [SL]	8. CHANNELS OF BEHAVIOR [CH]
• A spike in incoming loan applications during peak periods. • Rising non-performing assets prompting a review of the credit evaluation process. • Need to reduce turnaround time to stay competitive with other lenders.	An interactive, web-based Machine Learning application (built with Flask, scikit-learn, and XGBoost) that allows credit officers to input applicant details (income, loan amount, credit history, employment, etc.) to instantly predict loan approval likelihood, complete with fast-track and high-risk flagging.	ONLINE: Bank's internal web portal accessing the Flask application; batch upload for bulk applicant evaluation. OFFLINE: Branch loan-approval desks and credit risk review meetings.

4. EMOTIONS BEFORE / AFTER [EM]
Before: Overwhelmed by high application volumes, anxious about approving risky applicants and increasing NPAs, frustrated by the time-consuming nature of manual document review. After: Confident in data-driven approval decisions, relieved by fast and consistent automated predictions, empowered to fast-track low-risk applicants and flag high-risk ones for scrutiny.